



The Inflammatory Microcytosis Paradox: What Does HbA2 Really Tell Us?

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INTRODUCTION

Microcytosis is commonly encountered in routine hematology and is classically associated with iron deficiency and thalassemia. However, inflammatory and infectious disorders may also produce microcytosis through impaired iron availability for erythropoiesis. Interpretation becomes challenging when a patient presents with marked microcytosis, apparently normal iron stores and a non-contributive hemoglobin electrophoretic profile. A low HbA2 may be observed in iron deficiency and α -thalassemia and may therefore complicate the interpretation of a microcytic profile. Recent studies emphasize that reduced HbA2 has multiple potential causes and should not be interpreted in isolation. In our laboratory, we repeatedly observed patients with microcytosis, normal or apparently preserved ferritin and an inflammatory context, sometimes associated with a low HbA2 level. These observations prompted the present prospective study.

METHODS END MATERIALS

Prospective, observational, analytical study conducted from January to July 2026.
Study population : Patients investigated for microcytosis/anemia who underwent hematological and hemoglobin studies during the study period .
Laboratory parameters: The following parameters were collected: Hemoglobin; Red blood cell count ; Mean corpuscular volume (MCV); Mean corpuscular hemoglobin (MCH); Platelet count ;Ferritin; C-reactive protein (CRP) ; Erythrocyte sedimentation rate (ESR) Hemoglobin electrophoresis: HbA, HbA2 and HbF when available. Analytical approach Patients were characterized according to inflammatory status and HbA2 profile. Descriptive statistics were performed on the cases currently available. Correlation analyses between inflammatory markers (CRP and ESR) , MCV and HbA2 will be performed after completion of the full cohort.

RESULTS AND DISCUSSION

Seven patients, including a pair of male twins, were included in this preliminary series. All documented cases showed microcytosis and hypochromia, with MCV ranging from 53.6 to 73.0 fL and MCH from 16.5 to 23.4 pg. Hemoglobin levels ranged from 6.7 to 12.4 g/dL. Ferritin was normal or elevated in all patients with available results (14.44–375), despite inflammatory or infectious contexts in most cases. Positive CRP and/or elevated ESR were documented in patients with persistent nosocomial infection, inflammatory syndrome, chronic sarcoidosis and brucellosis. Hemoglobin electrophoresis was globally non-contributive, with HbA2 values ranging from 1.6% to 3.0%. Reduced HbA2 values (1.6–1.8%) were observed in some patients, whereas normal HbA2 values were found despite active inflammation in others, including a patient with brucellosis (HbA2 3.0%). This heterogeneity does not support a systematic relationship between inflammation and HbA2 reduction.
The twin pair represented a particularly informative familial observation: both 4-year-old boys exhibited microcytosis and hypochromia with relatively preserved RBC counts and normal hemoglobin electrophoretic profiles, while their parents and three siblings had normal CBC and electrophoretic profiles. This familial pattern raises the possibility of a genetic microcytic trait, particularly α -thalassemia, which may remain undetected by conventional electrophoresis. Indeed, α -thalassemia trait typically presents with microcytosis and normal hemoglobin electrophoresis.
Overall, these findings highlight the diagnostic complexity of microcytosis in inflammatory settings. Normal or elevated ferritin does not necessarily reflect adequate iron availability during inflammation, while a normal electrophoretic profile does not exclude an underlying globin disorder.

TABLE 2: HEMOGLOBIN ELECTROPHORESIS

Patient	Clinical / Inflammatory context	HbA (%)	HbA2 (%)	HbF (%)	Interpretation
P1	Inflammatory syndrome	97.6	3.3	/	Normal profile
P2	Persistent nosocomial infection	96.0	1.8	2.2	Reduced HbA2
P3	Marked inflammatory syndrome	97.5	2.7	/	Normal
P4	Chronic sarcoidosis	97.7	2.3	/	Normal
P5 – Twin	Inflammatory syndrome	97.6	2.4	/	Normal profile
P6	HTA, type 2 diabetes, hypothyroidism	98.4	1.6	/	Reduced HbA2
P7	Brucellosis + inflammatory syndrome	97.0	3.0		Normal

CONCLUSION

Microcytosis associated with inflammation represents a diagnostic challenge in hemoglobin electrophoresis interpretation. Low HbA2 may occur in some inflammatory conditions, but this association appears heterogeneous rather than systematic. A normal electrophoretic profile should not exclude further investigation, particularly in patients with persistent microcytosis and normal ferritin. An integrated interpretation of inflammatory markers, iron parameters and erythrocyte indices may improve the diagnostic approach and help identify patients requiring additional molecular investigations.

Bibliographies

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OBJECTIVES

- To investigate the relationship between inflammatory status, erythrocyte indices and hemoglobin electrophoretic parameters in patients presenting microcytosis, with particular emphasis on HbA2.
- To determine the frequency of reduced HbA2 among microcytic patients with inflammatory conditions.
- To assess the relationship between HbA2 and MCV.
- To evaluate whether a normal ferritin value reliably excludes iron-related mechanisms in inflammatory states.
- To identify microcytic profiles in which a normal electrophoresis may conceal an underlying hemoglobinopathy, particularly α -thalassemia.

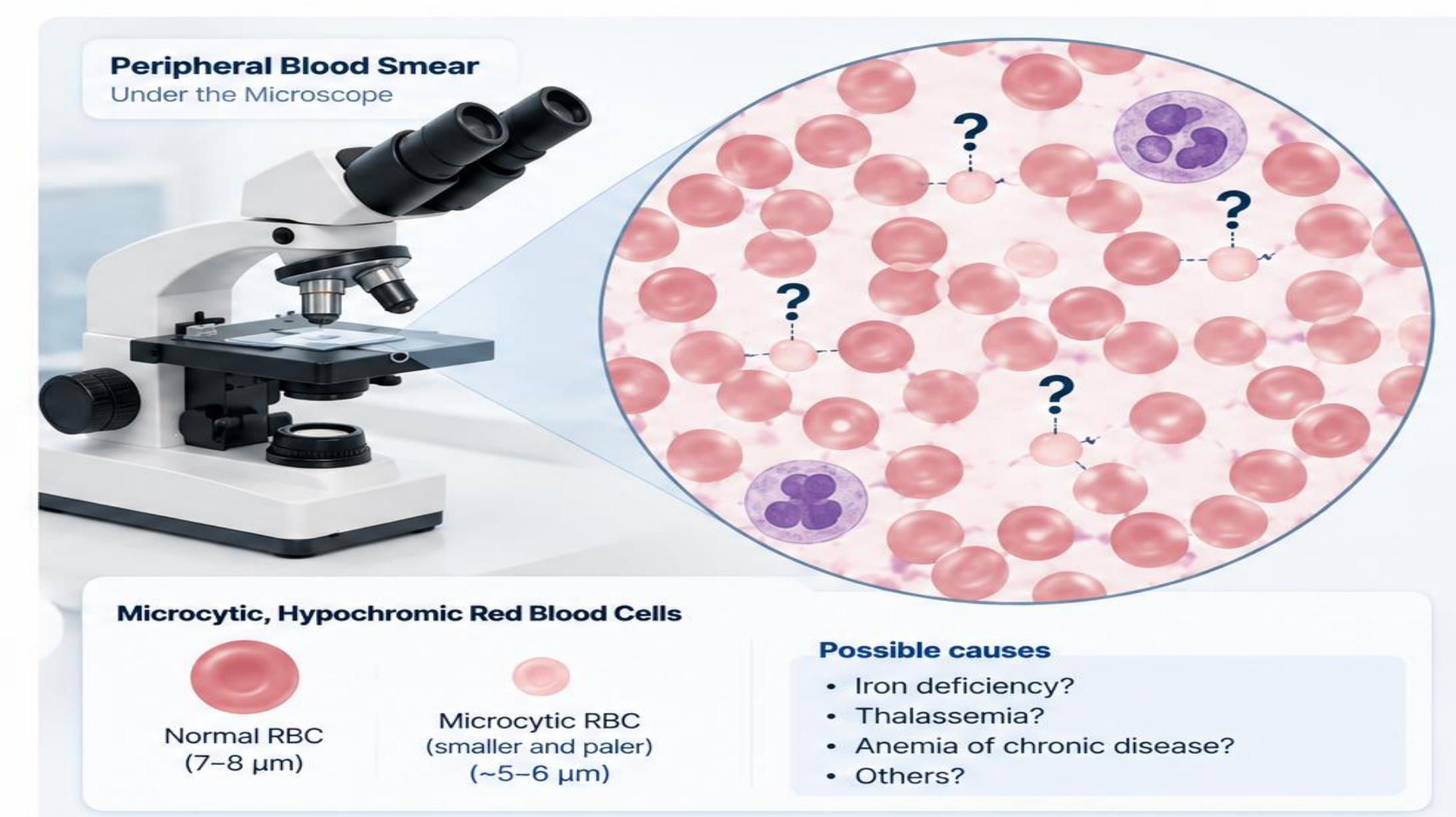


TABLE 1: CLINICAL AND BIOLOGICAL CHARACTERISTICS

Patient	Sex / Age	Clinical context	RBC (G/L)	Hb (g/dL)	MCV (fL)	MCH (pg)	Platelets (G/L)	Ferritin	CRP	ESR (mm/h)
P1	Male / 4 y	Inflammatory syndrome; microcytic hypochromic anemia	5.42	11.1	67.3	20.4	300	29.48	Positive	32 ↑
P2	Male / 9 mo	Prematurity; prolonged hospitalization; persistent nosocomial infection	4.006	6.7	53.6	16.5	599	31	Positive	NR
P3	Female / 8y	Marked inflammatory syndrome; microcytic hypochromic anemia	4.54	8.7	59.7	19.2	180	60	Strongly positive	90 ↑
P4	Male / 18 y	Chronic sarcoidosis for 8 years	5.12	11.62	72.5	22.1	200	67	NR	NR
P5 – Twin	Male / 4 y	Twin of P1; inflammatory syndrome; microcytic hypochromic anemia	5.43	12.1	71.8	22.4	290	14.44	NR	NR
P6	Male / 54 y	Hypertension, type 2 diabetes, hypothyroidism; microcytic hypochromic anemia	4.02	9.4	70.0	23.3	479	Normal	NR	NR
P7	Male / Adult	Brucellosis with inflammatory syndrome	5.30	12.4	73.0	23.4	NR	375 ↑	Positive	Accelerated; value

TABLE 3 : FAMILIAL INVESTIGATION OF THE TWIN PAIR

Family member	Sex	CBC / FNS	Ferritin	Hemoglobin electrophoresis	Interpretation
Father	Male	Normal	Normal	Normal	No abnormality reported
Mother	Female	Normal	Normal	Normal	No abnormality reported
P1 – Twin 1	Male	Microcytic / hypochromic	29.48	Normal	Microcytosis with normal electrophoresis
P5 – Twin 2	Male	Microcytic / hypochromic	14.44	Normal	Microcytosis with normal electrophoresis
3 brothers	Male	Normal	NR	Normal	No abnormality reported
1 sister	Female	Normal	NR	Normal	No abnormality reported